

GroupTrack

Quick Start Guide for Testers

Off-Road Convoy Tracking • Trail Maps • GPS Recording

App	GroupTrack (closed beta)
Platform	Android 10 or newer
Radios	Meshtastic T1000-E (optional — see below)
Website	grouptrack.org

Two Ways to Ride

	Solo Mode (Phone Only)	Group Mode (With Radios)
What you need	Just your Android phone	Phone + Meshtastic radio per rider
Radios required	None	3–4 recommended for full experience
What you get	GPS recording, trail maps, map downloads, track viewing	Everything in solo mode PLUS real-time convoy tracking, lead/follow
Lead cart	Automatic — you are the only rider	Select lead cart before starting the ride
Cell service	Not needed on trail (download maps before)	Not needed — radios communicate via mesh network
Best for	Exploring, solo rides, evaluating the app	Group rides, convoy coordination

1. First Launch — Permissions

On first launch, GroupTrack requests several permissions. Grant all of these for the app to function properly:

Permission	What to Select	Why
Location	Allow all the time	GPS tracking, trail display, map centering. Must be "all the time" for background.
Nearby Devices	Allow	Connecting to Meshtastic radios over Bluetooth.
Bluetooth	Allow	BLE communication with radios.
Notifications	Allow	Foreground service alerts during GPS recording and downloads.
Files / Storage	Allow (if prompted)	Map tiles and trail data stored in shared storage.

Accidentally denied a permission? Go to Settings → Apps → GroupTrack → Permissions and change it to Allow.

2. Pre-Ride Setup — Do This at Home on WiFi

Everything in this section happens before you leave the house. WiFi is required for map downloads. Radio programming requires a connected radio.

2A. Download Maps

GroupTrack stores map tiles locally for offline use on the trail. Download before you ride.

Step 1. Open GroupTrack. The **Convoy Map** loads by default. You can also use the **Map Viewer** from the menu — same download controls, no convoy overlays.

Step 2. Pan and zoom to the area you plan to ride.

Step 3. Tap the **menu (≡)** → **Download Maps**.

Step 4. Select your tile source:

Source	Best For
SAT	Satellite imagery — terrain, roads, obstacles
TOPO	Topographic contour lines — elevation and shape
HYB	Satellite + trail overlays — best of both
TOPO+	Enhanced topo — detailed contours with shading

Step 5. Select zoom levels. **z14** for a wide area overview (low storage). **z15–18** for detailed ride area.

Step 6. Tap **Download**. Progress shows in the notification bar.

Tip: Download z14 for the whole region first. Then download z15–18 for just the trail system you plan to ride. This minimizes storage while giving you the detail you need.

Both Convoy Map and Map Viewer share the same tile cache. Download from either one.

2B. Program Your Radio (Group Mode Only)

Skip this section if riding solo (phone only). For group rides, each radio in the convoy should run the GroupTrack convoy config.

Step 1. Power on your Meshtastic radio (T1000-E recommended).

Step 2. In GroupTrack, tap the **Bluetooth icon** in the top bar.

Step 3. Your radio appears in the device list. Tap to connect.

Step 4. Once connected, tap **menu (≡) → Radio Configuration**.

Step 5. Tap **Apply Convoy Config**. This writes the off-road optimized profile.

Step 6. The radio reboots with new settings. Wait for reconnect.

Step 7. Repeat for each radio in the group.

Setting	Value	Why
Modem Preset	Long Range / Slow	Maximum range in canyons and mountains
TX Power	Maximum (30 dBm)	Penetrate terrain and tree cover
Position Interval	3 seconds	Smooth real-time convoy tracking
Region	US (902–928 MHz)	FCC compliant

Your original radio config is automatically archived. Restore it anytime from the Radio Configuration screen.

All radios in the convoy should run the same config for best mesh performance.

2C. Explore Trails

GroupTrack includes over 5,700 trails from the Utah Geospatial Resource Center covering southern Utah — Sand Hollow, Hurricane Cliffs, Warner Valley, Paiute Trail, and more.

Step 1. Open either the **Convoy Map** or **Map Viewer**.

Step 2. Pan to southern Utah.

Step 3. Tap the **TRAILS** button to toggle trail overlay on/off.

Step 4. Trails appear as colored lines. Zoom in to see trail detail.

Trail data works fully offline after first load. Coverage is expanding — USFS, BLM, and additional states coming soon.

3. On the Trail — Ride Experience

3A. Solo Mode (Phone Only — No Radio)

No radio needed. Your phone is the tracker. GroupTrack uses your phone's GPS to record your ride and display your position on the map.

Starting a ride:

Step 1. On the Convoy Map, tap the **Record (●) button**.

Step 2. Recording starts immediately. No lead cart selection needed — you are the only rider and lead is assigned automatically.

Step 3. The record button turns red. Your position shows on the map as a single cart icon.

Step 4. Ride! Your track draws on the map in real time.

During the ride:

- Your position updates on the map as you move
- Downloaded map tiles display even with no cell service
- Toggle TRAILS to see trail overlays on the map
- The map auto-pans to follow your position

Stopping a ride:

Step 1. Tap the **Stop (■) button**.

Step 2. Enter a name for your track when prompted.

Step 3. Your GPX file is saved. View it later in the Map Viewer.

Solo mode is great for exploring, scouting new trails, and evaluating GroupTrack before investing in radios.

3B. Group Mode (3–4 Radios — Full Convoy Tracking)

This is where GroupTrack shines. Every rider sees every other rider's position in real time — no cell service required. The mesh radios communicate directly with each other.

Starting a group ride:

Step 1. All riders: power on radios, open GroupTrack, confirm radio is connected (Bluetooth icon shows connected state).

Step 2. Riders should appear as cart icons on each other's maps.

Step 3. One rider taps the **Record (●) button**.

Step 4. A dialog appears: **select the lead cart**. This is the rider at the front of the convoy. Tap their callsign.

Step 5. Recording starts. All riders are visible on the map.

What you see during a group ride:

- **Cart icons** for every rider with their callsign label
- **Lead cart** highlighted at the front of the convoy
- **Convoy spread** — distance between the lead and tail riders
- **Speed and heading** for each rider
- **Real-time track** colored by which cart passed through each segment
- **Position updates every 3 seconds** via mesh radio

Best practices:

- Assign lead cart before the group starts moving
- All radios should run the convoy config (Section 2B)
- Keep radios powered throughout the ride — BLE drops if a radio powers off
- Start GPS recording before you move

Even two radios are useful — you see each other and the distance between you. But 3–4 radios give the full convoy experience with lead, middle, and tail positions.

4. After the Ride — Viewing Tracks

Step 1. Open **Map Viewer** from the menu.

Step 2. Tap the **track picker** to browse your saved tracks.

Step 3. Select one or more tracks to display on the map.

Step 4. Tracks render as colored lines over your downloaded map tiles.

You can import GPX/KML files from other apps using the file picker. Great for comparing your ride against planned routes.

5. Troubleshooting

Problem	Solution
Radio not showing in scan	Toggle Bluetooth off/on in Android settings. Power cycle the radio. A radio can only connect to one device at a time.
Map is blank or gray	You need map tiles. Connect to WiFi and download tiles for your area (Section 2A).
No GPS position on map	Check that Location is set to "Allow all the time" in Settings → Apps → GroupTrack → Permissions.
App shows ANR dialog	Tap "Wait" — the app will load. First launch is slower. Report to the GroupTrack team if this happens often.
Trails not showing	Tap the TRAILS button to toggle visibility. Pan to southern Utah where trail data is currently available.
Track not recording	Confirm the record button is red (active). Verify background location permission is granted.
Radio keeps disconnecting	Some budget Android devices have BLE compatibility issues. Contact us for the compatible device list.
Other riders not visible	Confirm all radios are running the same convoy config. Check that all radios show as connected in the app.

6. Providing Feedback

Your feedback shapes GroupTrack. As a closed beta tester, you are directly influencing the app's development.

- Report bugs with as much detail as possible — what you were doing, what happened, what device you're on
- Tell us what works well and what feels awkward
- Screenshots or screen recordings are extremely helpful
- Feature suggestions are welcome

Thank you for testing GroupTrack. Your time on the trail is building the app.

